

Amechi Aduba

(914) 837-0883 • amechia99@gmail.com • Cortlandt Manor, NY • GitHub • LinkedIn • Portfolio

PROFESSIONAL SUMMARY

Software engineer and computer science graduate with experience designing, implementing, testing, and maintaining data-driven applications across healthcare, financial services, and research environments. Hands-on background in Java, Python, SQL, database systems, applications programming, automated validation, and technical documentation. Proven ability to own complex projects from requirements and systems analysis through deployment, communicate risks clearly, and improve reliability, scalability, and data quality.

EDUCATION

University of Pennsylvania, Online

Expected May 2028

M.S.E in Artificial Intelligence

Clark University, Worcester, MA

May 2026

B.A. in Computer Science | Minor in Data Science | Dean's List & Honors: Fall 2023, Fall 2024

Relevant Coursework: AI (Graduate) • Computer Networks (Graduate) • Machine Learning • Web Development • Mobile Software Development

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, SQL, HTML/CSS

Systems & Software Engineering: Systems analysis, applications programming, software design, unit testing, quality assurance, troubleshooting, technical documentation

Databases & Data: MySQL, Oracle, MongoDB, NoSQL, database concepts, SQL query optimization, validation pipelines, data analysis

Applications & APIs: React, FastAPI, Node.js, Django, Supabase, RESTful APIs, microservices concepts

Cloud, Networking & DevOps: Azure, cloud-native concepts, Docker fundamentals, CI/CD basics, computer networks coursework, Git, GitHub

PROFESSIONAL EXPERIENCE

Pentiction Pediatric — Software & Web Developer

June 2026 – Current

- Led a scheduling application from requirements and workflow analysis through implementation, deployment, and maintenance for patients and therapists, serving as the primary contributor on a complex operational project.
- Implemented automated Twilio and Resend messaging, documented system workflows, and resolved production issues to improve reliability and reduce repetitive administrative work.

Clark University — Research Assistant, Computer Science Department

June 2025 – May 2026

- Designed, implemented, and validated a Java version of the Hierarchical Virtual Bitmap Estimator, applying unit tests and quality checks to assess correctness, scalability, and edge cases.
- Built reproducible Java and Python data-generation and analysis pipelines to evaluate system behavior at scale and isolate performance or accuracy issues.
- Produced graphical performance analyses and technical documentation that translated complex systems findings into clear decisions for faculty stakeholders.

Helius Health AI — Web Development & Backend Intern

November 2024 – September 2025

- Developed and tested a React and TypeScript application backed by Supabase, maintaining UI components and RESTful data flows while troubleshooting application and database issues.
- Implemented Python and Twilio backend automation, investigated integration failures, and communicated technical findings during rapid product iterations.

Cantor Fitzgerald / BGC — Data Science Intern

June 2024 – August 2024

- Designed and deployed an AI-driven automation tool for corporate actions processing, acting as a major contributor from systems analysis and requirements gathering through validation and presentation.
- Optimized complex Oracle SQL queries and automated validation scripts supporting live financial data feeds, strengthening data quality and application reliability.

PROJECTS

Footee Vision: Computer-Vision Player Analytics Platform

June 2026

- Built a multi-stage FastAPI, OpenCV, PyTorch, and React system that processes soccer footage through detection, tracking, and classification services.
- Analyzed system limitations, documented failure cases, and made performance and design tradeoffs for unreliable real-world video inputs.

Meechify: AI Music Recommendation Platform

Jan 2026

- Developed a full-stack FastAPI and React/Vite application with Spotify OAuth, custom ranking logic, and an OpenAI-compatible LLM API.
- Implemented API response handling and fallback behavior to reliably transform raw listening data into structured, user-facing insights.

Hand Tracking Sketch Recognition: AI Game

Dec 2025

- Built and tested a real-time webcam application using MediaPipe and a KNN classifier, troubleshooting browser, camera, canvas, and model-integration issues.

Ballon d'Or Nominee Analysis: Predictive Analytics Dashboard

Dec 2024

- Built a Python, Pandas, and Tableau analytics workflow that transformed player data into position-weighted rankings and interactive reports for non-technical users.

ACTIVITIES

Clark Varsity Men's Soccer • Clark BSU • ColorStack • MOCA (Men of Color Alliance) • Hack Diversity